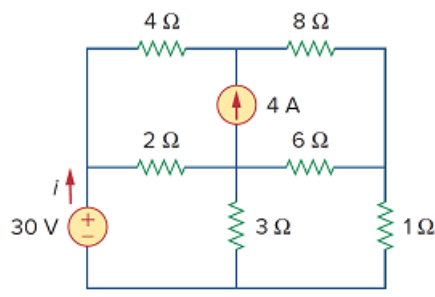


Problem

Find current i in the circuit of Fig. 3.91.

Figure 3.91:



Step-by-step solution

Step 1 of 8

Refer to figure 3.91 given in the textbook.

Redraw the circuit with loop currents as shown in figure 1.

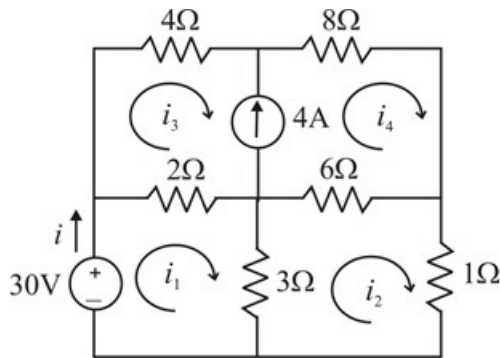


Figure 1

Step 2 of 8

Apply Mesh analysis to loop 1 of the circuit shown in figure 1.

$$-30 + 2(i_1 - i_3) + 3(i_1 - i_2) = 0$$

$$5i_1 - 3i_2 - 2i_3 = 30 \quad \dots\dots (1)$$

Apply Mesh analysis to loop 2 of the circuit shown in figure 1.

$$3(i_2 - i_1) + 6(i_2 - i_4) + 1(i_2) = 0$$

$$-3i_1 + 10i_2 - 6i_4 = 0 \quad \dots\dots (2)$$

There's a current source in between loops 1 and 2.

The super-mesh equation is,

$$i_4 - i_3 = 4 \quad \dots\dots (3)$$

Step 3 of 8

Redraw the circuit with the super-mesh as shown in figure 2.

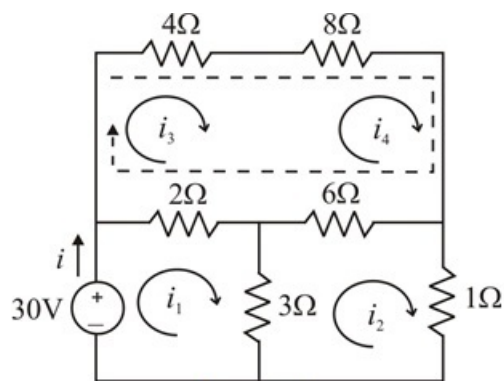


Figure 2

Step 4 of 8

Apply Kirchhoff's voltage law to super mesh shown in figure 2.

$$4(i_3) + 8(i_4) + 6(i_4 - i_2) + 2(i_3 - i_1) = 0$$

$$-2i_1 - 6i_2 + 6i_3 + 14i_4 = 0 \dots\dots (4)$$

Recall equation (3).

$$i_4 - i_3 = 4$$

$$i_4 = i_3 + 4$$

Recall equation (2).

$$-3i_1 + 10i_2 - 6i_4 = 0$$

Substitute $(i_3 + 4)$ for i_4 .

$$-3i_1 + 10i_2 - 6(i_3 + 4) = 0$$

$$-3i_1 + 10i_2 - 6i_3 - 24 = 0$$

$$-3i_1 + 10i_2 - 6i_3 = 24 \dots\dots (5)$$

Step 5 of 8

Recall equation (4).

$$-2i_1 - 6i_2 + 6i_3 + 14i_4 = 0$$

Substitute $(i_3 + 4)$ for i_4 .

$$-2i_1 - 6i_2 + 6i_3 + 14(i_3 + 4) = 0$$

$$-2i_1 - 6i_2 + 6i_3 + 14i_3 + 56 = 0$$

$$-2i_1 - 6i_2 + 20i_3 = -56 \dots\dots (6)$$

Step 6 of 8

Recall equation (5).

$$-3i_1 + 10i_2 - 6i_3 = 24$$

$$i_3 = \frac{-3i_1 + 10i_2 - 24}{6} \dots\dots(7)$$

Recall equation (6).

$$-2i_1 - 6i_2 + 20i_3 = -56$$

Consider equation (7).

$$-2i_1 - 6i_2 + 20\left(\frac{-3i_1 + 10i_2 - 24}{6}\right) = -56$$

$$-12i_1 - 36i_2 - 60i_1 + 200i_2 - 480 = -336$$

$$-72i_1 + 164i_2 = 144 \dots\dots (8)$$

Step 7 of 8

Recall equation (1).

$$5i_1 - 3i_2 - 2i_3 = 30$$

Substitute $\frac{-3i_1 + 10i_2 - 24}{6}$ for i_3 .

$$5i_1 - 3i_2 - 2\left(\frac{-3i_1 + 10i_2 - 24}{6}\right) = 30$$

$$30i_1 - 18i_2 + 6i_1 - 20i_2 + 48 = 180$$

$$36i_1 - 38i_2 = 132 \dots\dots (9)$$

Step 8 of 8

Recall equation (8).

$$-72i_1 + 164i_2 = 144$$

$$i_2 = \frac{144 + 72i_1}{164} \dots\dots (10)$$

Recall equation (9).

$$36i_1 - 38i_2 = 132$$

Consider equation (10).

$$36i_1 - 38\left(\frac{144 + 72i_1}{164}\right) = 132$$

$$5904i_1 - 5472 - 2736i_1 = 21648$$

$$3168i_1 = 27120$$

$$i_1 = \frac{27120}{3168}$$

$$i_1 = 8.5606$$

$$i_1 = 8.561\text{A}$$

The loop current i_1 is equal to current i shown in figure 2.

$$i = i_1$$

Substitute 8.561 A for i_1 .

$$i = 8.561\text{ A}$$

Therefore, the value of current i is 8.561 A.